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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 27th July Evening Shift

JEE Main Question Paper with Answer Key

27 Jul 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	22 questions
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Out of syllabus	19 questions
Answer key	at the end

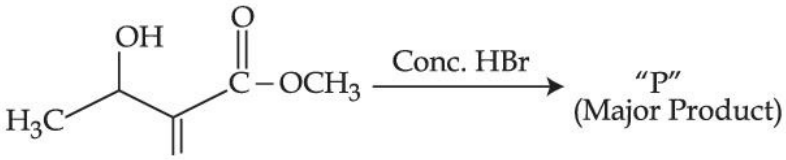
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Chemistry 22 questions

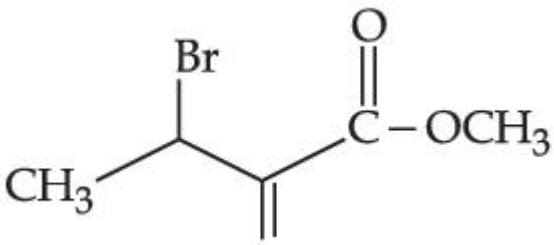
Q1. MCQ (Single Correct Answer)



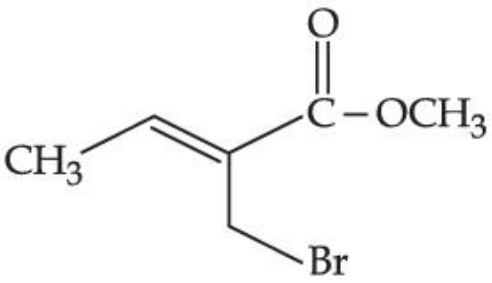
Open solution

Consider the above reaction, the major product "P" formed is :

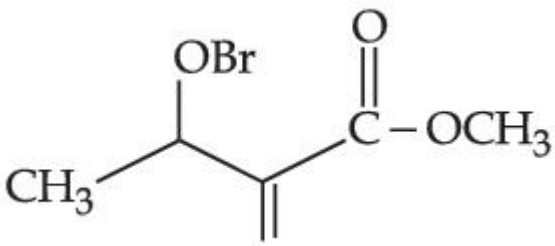
(A)



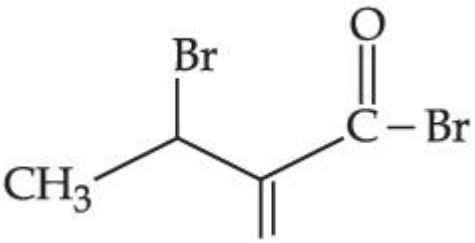
(B)



(C)



(D)



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Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : $SO_2(g)$ is adsorbed to a larger extent than $H_2(g)$ on activated charcoal.

Reason R : $SO_2(g)$ has a higher critical temperature than $H_2(g)$.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both A and R are correct but R is not the correct explanation of A.

(B) Both A and R are correct and R is the correct explanation of A.

(C) A is not correct but R is correct.

(D) A is correct but R is not correct.

Q3. MCQ (Single Correct Answer)

Open solution

The correct order of first ionisation enthalpy is :

(A) Mg < S < Al < P **(B)** Mg < Al < S < P **(C)** Al < Mg < S < P **(D)** Mg < Al < P < S

Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Hyperconjugation is a permanent effect.

Statement II : Hyperconjugation in ethyl cation $\left(CH_3-\overset{+}{C}H_2\right)$ involves the overlapping of $C_{sp^2}-H_{1s}$ bond with empty 2p orbital of other carbon.

Choose the correct option :

(A) Both statement I and statement II are false

(B) Statement I is incorrect but statement II is true

(C) Statement I is correct but statement II is false

(D) Both statement I and statement II are true.

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Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ are d^2sp^3 hybridised.

Statement II : $[MnCl_6]^{3-}$ and $[FeF_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons, respectively.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is correct but statement II is false

(B) Both statement I and Statement II are false

(C) Statement I is incorrect but statement II is true

(D) Both statement I and statement II are true

Q6. MCQ (Single Correct Answer)

Open solution

To an aqueous solution containing ions such as Al^{3+}, Zn^{2+}, Ca^{2+}, Fe^{3+}, Ni^{2+}, Ba^{2+} and Cu^{2+} was added conc. HCl, followed by H_2S.

The total number of cations precipitated during this reaction is/are :

(A) 1 (B) 3 (C) 4 (D) 2

Q7. MCQ (Single Correct Answer)

Open solution

Compound A gives D-Galactose and D-Glucose on hydrolysis. The compound A is :

(A) Amylose (B) Sucrose (C) Maltose (D) Lactose

Q8. MCQ (Single Correct Answer)

Open solution

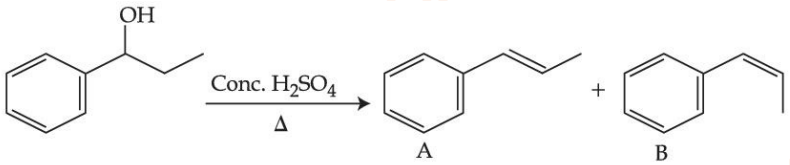
$$R-CN \xrightarrow[(ii)H_2O]{(i)DIBAL-H} R-Y$$

Consider the above reaction and identify "Y"

(A) $-CH_2NH_2$ (B) $-CONH_2$ (C) $-CHO$ (D) $-COOH$

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Q9. MCQ (Single Correct Answer)



Open solution

Consider the above reaction, and choose the correct statement :

(A) The reaction is not possible in acidic medium

(B) Both compounds A and B are formed equally

(C) Compound A will be the major product

(D) Compound B will be the major product

Q10. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List - I (compound)		**List - II (effect/affected species)**
(a)	Carbon monoxide	(i)	Carcinogenic
(b)	Sulphur dioxide	(ii)	Metabolized by pyrus plants
(c)	Polychlorinated biphenyls	(iii)	haemoglobin
(d)	Oxides of Nitrogen	(iv)	Stiffness of flower buds

Choose the correct answer from the options given below :

(A) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(B) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

(C) (a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

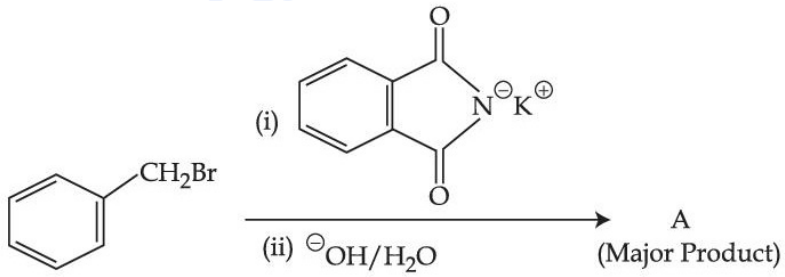
(D) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

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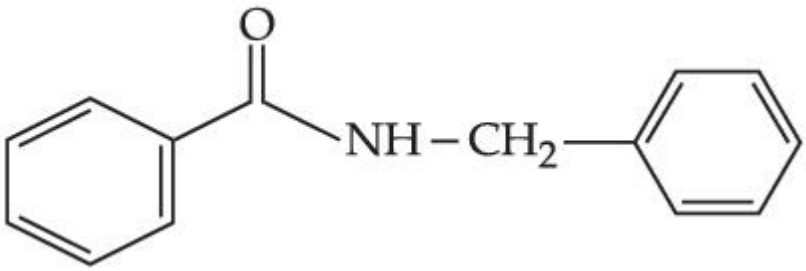
Q11. MCQ (Single Correct Answer)

Open solution

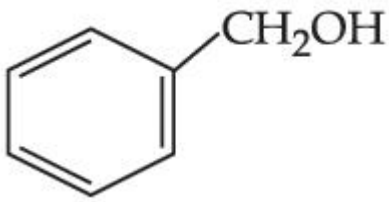
What is A in the following reaction?



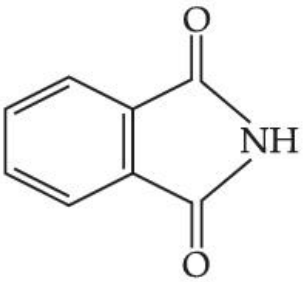
(A)



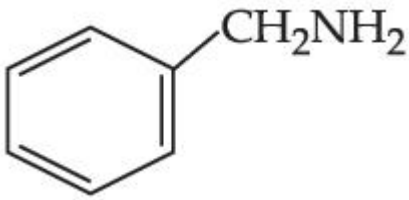
(B)



(C)



(D)

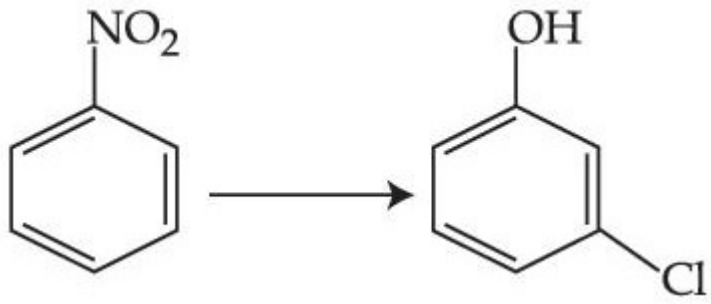


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Q12. MCQ (Single Correct Answer)

Open solution

The correct sequence of correct reagents for the following transformation is :



(A) (i) Fe, HCl

(ii) Cl_2, HCl

(iii) $NaNO_2$, HCl, 0°C

(iv) H_2O/H^+

(B) (i) Fe, HCl

(ii) $NaNO_2$, HCl, 0°C

(iii) H_2O/H^+

(iv) Cl_2, $FeCl_3$

(C) (i) Cl_2, $FeCl_3$

(ii) Fe, HCl

(iii) $NaNO_2$, HCl, 0°C

(iv) H_2O/H^+

(D) (i) Cl_2, $FeCl_3$

(ii) $NaNO_2$, HCl, 0°C

(iii) Fe, HCl

(iv) H_2O/H^+

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Q13. Numerical

The equilibrium constant for the reaction

$A(s) \rightleftharpoons M(s) + \frac{1}{2}O_2(g)$

is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. (Round off to the nearest integer)

Open solution

Q14. Numerical

When 400 mL of 0.2 M H_2SO_4 solution is mixed with 600 mL of 0.1 M NaOH solution, the increase in temperature of the final solution is _____ $\times 10^{-2}$ K. (Round off to the nearest integer).

[Use : $H^+(aq) + OH^-(aq) \rightarrow H_2O : \Delta_\gamma H = -57.1$ kJ mol^{-1}]

Specific heat of H_2O = 4.18 J K^{-1} g^{-1}

density of H_2O = 1.0 g cm^{-3}

Assume no change in volume of solution on mixing.

Open solution

Q15. Numerical

$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

The above reaction is carried out in a vessel starting with partial pressure P_{SO_2} = 250 m bar, P_{O_2} = 750 m bar and P_{SO_3} = 0 bar. When the reaction is complete, the total pressure in the reaction vessel is _____ m bar. (Round off of the nearest integer).

Open solution

Q16. Numerical

10.0 mL of 0.05 M $KMnO_4$ solution was consumed in a titration with 10.0 mL of given oxalic acid dihydrate solution. The strength of given oxalic acid solution is _____ $\times 10^{-2}$ g/L.

(Round off to the nearest integer)

Open solution

Q17. Numerical

The total number of electrons in all bonding molecular orbitals of O_2^{2-} is _____.

(Round off to the nearest integer)

Open solution

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Q18. Numerical

3 moles of metal complex with formula $Co(en)_2Cl_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.

(Round off to the nearest integer)

Open solution

Q19. Numerical

In a solvent 50% of an acid HA dimerizes and the rest dissociates. The van't Hoff factor of the acid is _____ $\times 10^{-2}$.

(Round off to the nearest integer)

Open solution

Q20. Numerical

The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is _____ degree. (Round off to the nearest integer)

Open solution

Q21. Numerical

For the first order reaction A $\rightarrow$ 2B, 1 mole of reactant A gives 0.2 moles of B after 100 minutes. The half life of the reaction is _____ min.

[Use : ln 2 = 0.69, ln 10 = 2.3]

Properties of logarithms : $\ln x^y = y \ln x$;

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

(Round off to the nearest integer)

Open solution

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Q22. Numerical

For the cell

$Cu(s) \mid Cu^{2+}(aq)\ (0.1\ M) \parallel Ag^{+}(aq)\ (0.01\ M) \mid Ag(s)$

the cell potential E_1 = 0.3095 V

For the cell

$Cu(s) \mid Cu^{2+}(aq)\ (0.01\ M) \parallel Ag^{+}(aq)\ (0.001\ M) \mid Ag(s)$

the cell potential = _____ $\times 10^{-2}$ V. (Round off to the nearest integer).

[Use : $\frac{2.303RT}{F}$ = 0.059]

Open solution

Mathematics 22 questions

Q23. MCQ (Single Correct Answer)

The point P (a, b) undergoes the following three transformations successively :

(a) reflection about the line y = x.

(b) translation through 2 units along the positive direction of x-axis.

(c) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$, then the value of 2a + b is equal to :

(A) 13 (B) 9 (C) 5 (D) 7

Open solution

Q24. MCQ (Single Correct Answer)

A possible value of 'x', for which the ninth term in the expansion of $\left\{3^{\log_3\sqrt{25^{x-1}+7}} + 3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}\right\}^{10}$ in the increasing powers of $3^{\left(-\frac{1}{8}\right)\log_3(5^{x-1}+1)}$ is equal to 180, is :

(A) 0 (B) -1 (C) 2 (D) 1

Open solution

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Q25. MCQ (Single Correct Answer)

Let f : R $\to$ R be defined as $f(x+y)+f(x-y)=2f(x)f(y), f\left(\frac{1}{2}\right)=-1$. Then, the value of $\sum_{k=1}^{20}\frac{1}{\sin(k)\sin(k+f(k))}$ is equal to :

Open solution

(A) $\text{cosec}^2(21)\cos(20)\cos(2)$

(B) $\sec^2(1)\sec(21)\cos(20)$

(C) $\text{cosec}^2(1)\text{cosec}(21)\sin(20)$

(D) $\sec^2(21)\sin(20)\sin(2)$

Q26. MCQ (Single Correct Answer)

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z-2| \le 1\}$ and

$S_2 = \{z \in C : z(1+i) + \bar{z}(1-i) \ge 4\}$.

Then, the maximum value of $\left|z-\frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to :

Open solution

(A) $\frac{3+2\sqrt{2}}{4}$

(B) $\frac{5+2\sqrt{2}}{2}$

(C) $\frac{3+2\sqrt{2}}{2}$

(D) $\frac{5+2\sqrt{2}}{4}$

Q27. MCQ (Single Correct Answer)

If $\tan\left(\frac{\pi}{9}\right), x, \tan\left(\frac{7\pi}{18}\right)$ are in arithmetic progression and $\tan\left(\frac{\pi}{9}\right), y, \tan\left(\frac{5\pi}{18}\right)$ are also in arithmetic progression, then $|x-2y|$ is equal to :

Open solution

(A) 4

(B) 3

(C) 0

(D) 1

Q28. MCQ (Single Correct Answer)

Let the mean and variance of the frequency distribution

x :	$x_1=2$	$x_2=6$	$x_3=8$	$x_4=9$
f :	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be :

Open solution

(A) 4

(B) 5

(C) $\frac{17}{3}$

(D) $\frac{16}{3}$

Q29. MCQ (Single Correct Answer)

The area of the region bounded by $y - x = 2$ and $x^2 = y$ is equal to :

Open solution

(A) $\frac{16}{3}$

(B) $\frac{2}{3}$

(C) $\frac{9}{2}$

(D) $\frac{4}{3}$

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Q30. MCQ (Single Correct Answer)

Let y = y(x) be the solution of the differential

equation $(x - x^3)dy = (y + yx^2 - 3x^4)dx$, $x > 2$. If $y(3) = 3$, then $y(4)$ is equal to :

(A) 4 (B) 12 (C) 8 (D) 16

Open solution

Q31. MCQ (Single Correct Answer)

The value of

$\lim\limits_{x\to 0}\left(\frac{x}{\sqrt[8]{1-\sin x}-\sqrt[8]{1+\sin x}}\right)$ is equal to :

(A) 0 (B) 4 (C) -4 (D) -1

Open solution

Q32. MCQ (Single Correct Answer)

Two sides of a parallelogram are along the lines 4x + 5y = 0 and 7x + 2y = 0. If the equation of one of the diagonals of the parallelogram is 11x + 7y = 9, then other diagonal passes through the point :

(A) (1, 2) (B) (2, 2) (C) (2, 1) (D) (1, 3)

Open solution

Q33. MCQ (Single Correct Answer)

Let $\alpha = \max\limits_{x\in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$ and $\beta = \min\limits_{x\in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$. If $8x^2 + bx + c = 0$ is a quadratic equation whose roots are $\alpha^{1/5}$ and $\beta^{1/5}$, then the value of c − b is equal to :

(A) 42 (B) 47 (C) 43 (D) 50

Open solution

Q34. MCQ (Single Correct Answer)

Let $f : [0, \infty) \to [0, 3]$ be a function defined by

$$f(x) = \begin{cases} \max\{\sin t : 0 \le t \le x\}, & 0 \le x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$$

Then which of the following is true?

(A) f is continuous everywhere but not differentiable exactly at one point in (0, ∞)

(B) f is differentiable everywhere in (0, ∞)

(C) f is not continuous exactly at two points in (0, ∞)

(D) f is continuous everywhere but not differentiable exactly at two points in (0, ∞)

Open solution

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Q35. MCQ (Single Correct Answer)

Let N be the set of natural numbers and a relation R on N be defined by $R = \{(x, y) \in N \times N : x^3 - 3x^2y - xy^2 + 3y^3 = 0\}$. Then the relation R is :

(A) symmetric but neither reflexive nor transitive

(B) reflexive but neither symmetric nor transitive

(C) reflexive and symmetric, but not transitive

(D) an equivalence relation

Open solution

Q36. MCQ (Single Correct Answer)

Consider a circle C which touches the y-axis at (0, 6) and cuts off an intercept $6\sqrt{5}$ on the x-axis. Then the radius of the circle C is equal to :

(A) $\sqrt{53}$ (B) 9 (C) 8 (D) $\sqrt{82}$

Open solution

Q37. MCQ (Single Correct Answer)

Let f : (a, b) $\rightarrow$ R be twice differentiable function such that $f(x) = \int_a^x g(t)dt$ for a differentiable function g(x). If f(x) = 0 has exactly five distinct roots in (a, b), then g(x)g'(x) = 0 has at least :

(A) twelve roots in (a, b)

(B) five roots in (a, b)

(C) seven roots in (a, b)

(D) three roots in (a, b)

Open solution

Q38. Numerical

If the real part of the complex number $z = \frac{3+2i\cos\theta}{1-3i\cos\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2\theta$ is equal to _____.

Open solution

Q39. Numerical

If $\int_0^{\pi} (\sin^3 x)e^{-\sin^2 x}dx = \alpha - \frac{\beta}{e}\int_0^1 \sqrt{t}e^t dt$, then $\alpha + \beta$ is equal to _____.

Open solution

Q40. Numerical

The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^x + 1 = 0$ is equal to _____.

Open solution

Q41. Numerical

Let y = y(x) be the solution of the differential equation $dy = e^{\alpha x + y}dx$; $\alpha \in$ N. If $y(\log_e 2) = \log_e 2$ and $y(0) = \log_e\left(\frac{1}{2}\right)$, then the value of α is equal to _____.

Open solution

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Q42. Numerical

Let n be a non-negative integer. Then the number of divisors of the form "4n + 1" of the number $(10)^{10} \cdot (11)^{11} \cdot (13)^{13}$ is equal to _____.

Open solution

Q43. Numerical

Let A = {n ∈ N | $n^2 \leq$ n + 10,000}, B = {3k + 1 | k∈ N} and C = {2k | k∈N}, then the sum of all the elements of the set A ∩(B − C) is equal to _____.

Open solution

Q44. Numerical

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and M = A + A^2 + A^3 + ... + A^{20}, then the sum of all the elements of the matrix M is equal to _____.

Open solution

Physics 27 questions

Q45. MCQ (Single Correct Answer)

An electron and proton are separated by a large distance. The electron starts approaching the proton with energy 3 eV. The proton captures the electron and forms a hydrogen atom in second excited state. The resulting photon is incident on a photosensitive metal of threshold wavelength 4000$\mathring{A}$. What is the maximum kinetic energy of the emitted photoelectron?

Open solution

(A) 7.61 eV

(B) 1.41 eV

(C) 3.3 eV

(D) No photoelectron would be emitted

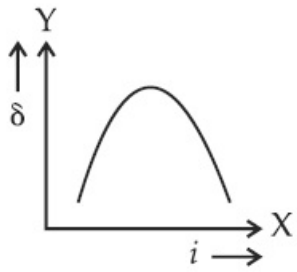
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Q46. MCQ (Single Correct Answer)

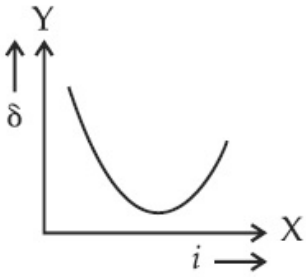
Open solution

The expected graphical representation of the variation of angle of deviation 'δ' with angle of incidence 'i' in a prism is :

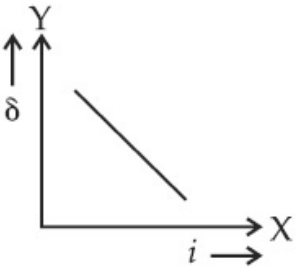
(A)



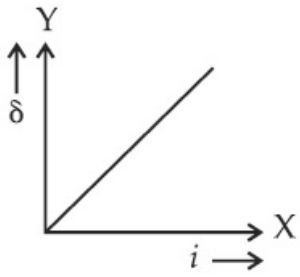
(B)



(C)



(D)



Q47. MCQ (Single Correct Answer)

Open solution

A raindrop with radius R = 0.2 mm falls from a cloud at a height h = 2000 m above the ground. Assume that the drop is spherical throughout its fall and the force of buoyance may be neglected, then the terminal speed attained by the raindrop is :

[Density of water f_w = 1000 kg m^{-3} and Density of air f_a = 1.2 kg m^{-3}, g = 10 m/s^2, Coefficient of viscosity of air = 1.8×10^{-5} Nsm^{-2}]

(A) 250.6 ms^{-1} (B) 43.56 ms^{-1} (C) 4.94 ms^{-1} (D) 14.4 ms^{-1}

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Q48. MCQ (Single Correct Answer)

Open solution

One mole of an ideal gas is taken through an adiabatic process where the temperature rises from 27° C to 37° C. If the ideal gas is composed of polyatomic molecule that has 4 vibrational modes, which of the following is true? [R = 8.314 J mol^{-1} K^{-1}]

(A) work done by the gas is close to 332 J

(B) work done on the gas is close to 582 J

(C) work done by the gas is close to 582 J

(D) work done on the gas is close to 332 J

Q49. MCQ (Single Correct Answer)

Open solution

An object of mass 0.5 kg is executing simple harmonic motion. It amplitude is 5 cm and time period (T) is 0.2 s. What will be the potential energy of the object at an instant $t = \frac{T}{4}s$ starting from mean position. Assume that the initial phase of the oscillation is zero.

(A) 0.62 J

(B) 6.2×10^{-3} J

(C) 1.2×10^{3} J

(D) 6.2×10^{3} J

Q50. MCQ (Single Correct Answer)

Open solution

Match List I with List II.

	List - I		List - II
(a)	Capacitance, C	(i)	$M^1L^1T^{-3}A^{-1}$
(b)	Permittivity of free space, ε_0	(ii)	$M^{-1}L^{-3}T^4A^2$
(c)	Permeability of free space, μ_0	(iii)	$M^{-1}L^{-2}T^4A^2$
(d)	Electric field, E	(iv)	$M^1L^1T^{-2}A^{-2}$

Choose the correct answer from the options given below

(A) (a) → (iii), (b) → (ii), (c) → (iv), (d) → (i)

(B) (a) → (iii), (b) → (iv), (c) → (ii), (d) → (i)

(C) (a) → (iv), (b) → (ii), (c) → (iii), (d) → (i)

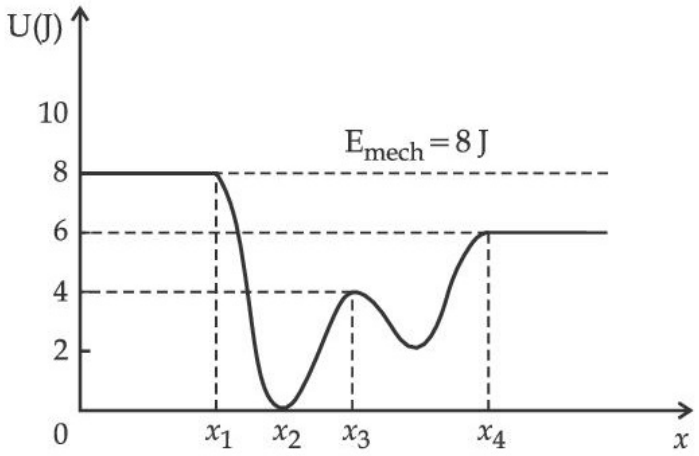
(D) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

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Q51. MCQ (Single Correct Answer)

Open solution

Given below is the plot of a potential energy function U(x) for a system, in which a particle is in one dimensional motion, while a conservative force F(x) acts on it. Suppose that E_{mech} = 8 J, the incorrect statement for this system is :



[where K.E. = kinetic energy]

(A) at $x > x_4$ K.E. is constant throughout the region.

(B) at $x < x_1$, K.E. is smallest and the particle is moving at the slowest speed.

(C) at $x = x_2$, K.E. is greatest and the particle is moving at the fastest speed.

(D) at $x = x_3$, K.E. = 4 J.

Q52. MCQ (Single Correct Answer)

Open solution

A 100Ω resistance, a 0.1 μF capacitor and an inductor are connected in series across a 250 V supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is 60 Hz.

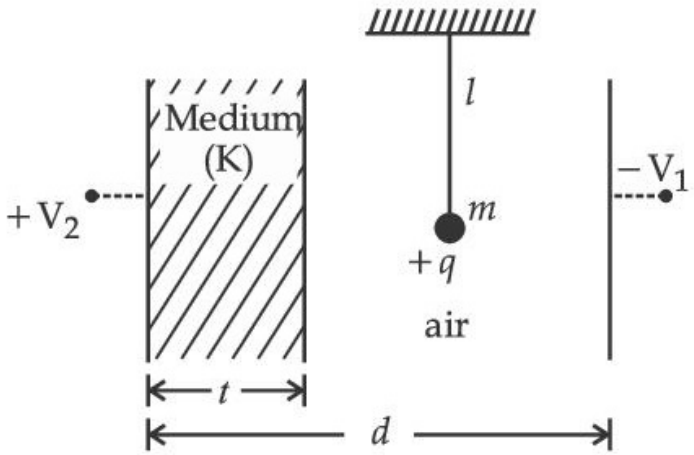
(A) 0.70 H **(B)** 70.3 mH **(C)** 7.03×10^{-5} H **(D)** 70.3 H

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Q53. MCQ (Single Correct Answer)

Open solution

A simple pendulum of mass 'm', length 'l' and charge '+ q' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be



(A) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

(B) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

(C) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

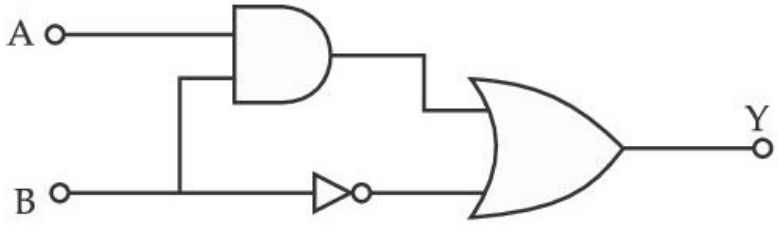
(D) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

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Q54. MCQ (Single Correct Answer)

Open solution

Find the truth table for the function Y of A and B represented in the following figure.



(A)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

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Q55. MCQ (Single Correct Answer)

Open solution

Figures A and B show two long straight wires of circular cross-section (a and b with a < b), carrying current I which is uniformly distributed across the cross-section. The magnitude of magnetic field B varies with radius r and can be represented as :

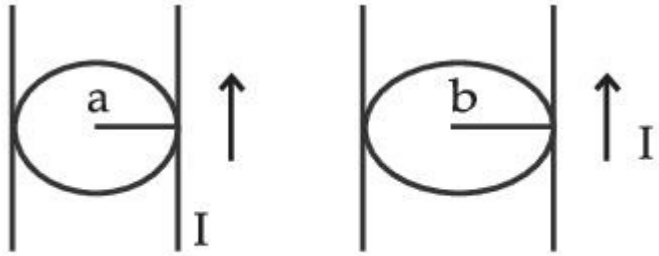
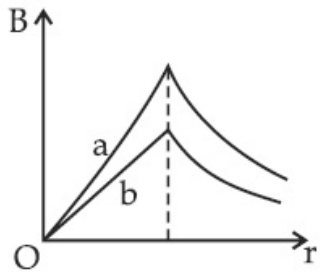
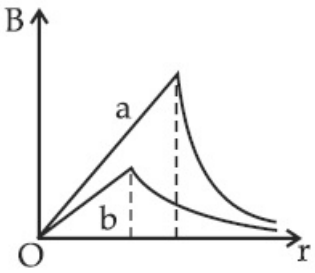


Fig. A Fig. B

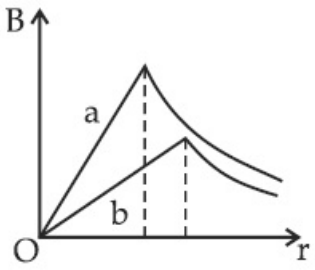
(A)



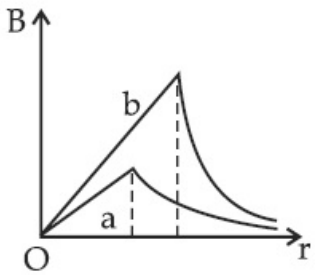
(B)



(C)



(D)



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Q56. MCQ (Single Correct Answer)

Two identical particles of mass 1 kg each go round a circle of radius R, under the action of their mutual gravitational attraction. The angular speed of each particle is :

(A) $\sqrt{\frac{G}{2R^3}}$

(B) $\frac{1}{2}\sqrt{\frac{G}{R^3}}$

(C) $\frac{1}{2R}\sqrt{\frac{1}{G}}$

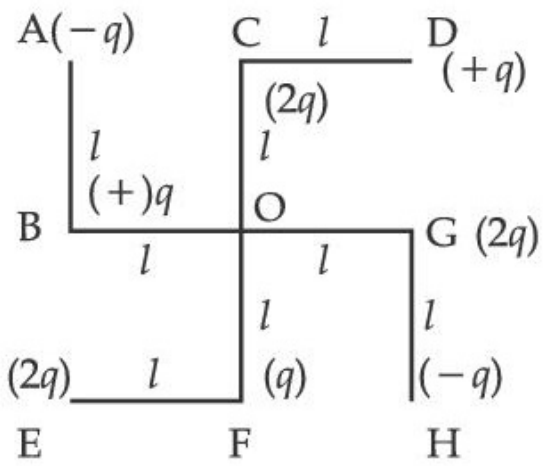
(D) $\frac{2G}{R^3}$

Open solution

Q57. MCQ (Single Correct Answer)

What will be the magnitude of electric field at point O as shown in the figure? Each side of the figure is l and perpendicular to each other.

Open solution



(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{(2l^2)}\left(2\sqrt{2}-1\right)$

(C) $\frac{q}{4\pi\varepsilon_0(2l)^2}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{2l^2}\left(\sqrt{2}\right)$

Q58. MCQ (Single Correct Answer)

A physical quantity 'y' is represented by the formula $y = m^2 r^{-4} g^x l^{-\frac{3}{2}}$

If the percentage errors found in y, m, r, l and g are 18, 1, 0.5, 4 and p respectively, then find the value of x and p.

Open solution

(A) 5 and ± 2

(B) 4 and ± 3

(C) $\frac{16}{3}$ and $\pm\frac{3}{2}$

(D) 8 and $\pm$ 2

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Q59. MCQ (Single Correct Answer)

An automobile of mass 'm' accelerates starting from origin and initially at rest, while the engine supplies constant power P. The position is given as a function of time by :

(A) $\left(\frac{9P}{8m}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$ (B) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}}t^{\frac{2}{3}}$

(C) $\left(\frac{9m}{8P}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$ (D) $\left(\frac{8P}{9m}\right)^{\frac{1}{2}}t^{\frac{3}{2}}$

Open solution

Q60. MCQ (Single Correct Answer)

The planet Mars has two moons, if one of them has a period 7 hours, 30 minutes and an orbital radius of 9.0×10^3 km. Find the mass of Mars.

$\left\{Given\ \frac{4\pi^2}{G} = 6 \times 10^{11} N^{-1} m^{-2} kg^2\right\}$

(A) 5.96×10^{19} kg (B) 3.25×10^{21} kg (C) 7.02×10^{25} kg (D) 6.00×10^{23} kg

Open solution

Q61. MCQ (Single Correct Answer)

A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$F = F_0\left[1 - \left(\frac{t-T}{T}\right)^2\right]$

Where F_0 and T are constants. The force acts only for the time interval 2T. The velocity v of the particle after time 2T is :

(A) $2F_0T/M$ (B) $F_0T/2M$ (C) $4F_0T/3M$ (D) $F_0T/3M$

Open solution

Q62. MCQ (Single Correct Answer)

The resistance of a conductor at 15°C is 16Ω and at 100°C is 20Ω. What will be the temperature coefficient of resistance of the conductor?

(A) $0.010°C^{-1}$ (B) $0.033°C^{-1}$ (C) $0.003°C^{-1}$ (D) $0.042°C^{-1}$

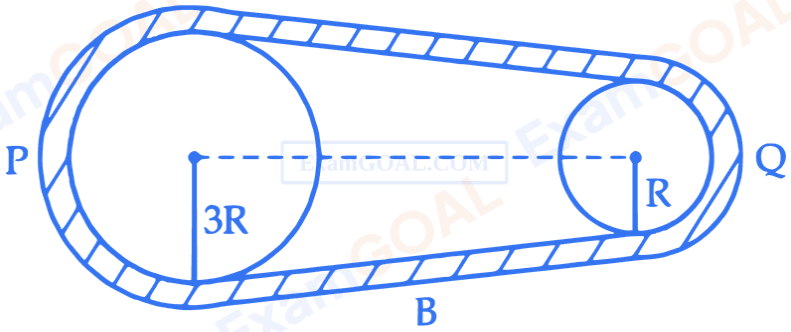
Open solution

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Q63. Numerical

In the given figure, two wheels P and Q are connected by a belt B. The radius of P is three times as that of Q. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be x : 1. The value of x will be _____.

Open solution



Q64. Numerical

The difference in the number of waves when yellow light propagates through air and vacuum columns of the same thickness is one. The thickness of the air column is _____ mm.
[Refractive index of air = 1.0003, wavelength of yellow light in vacuum = 6000 $\overset{o}{A}$]

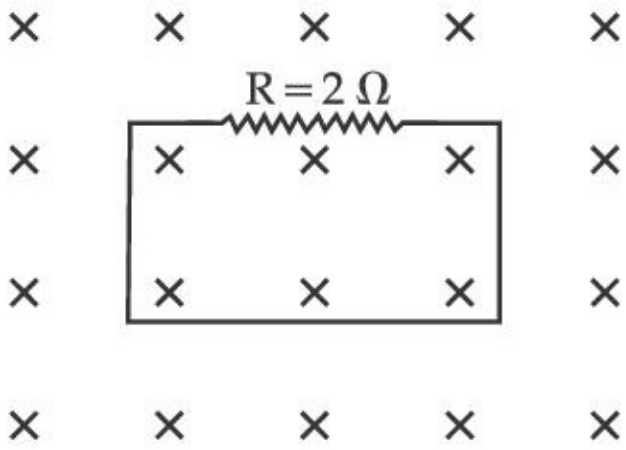
Open solution

Q65. Numerical

In the given figure the magnetic flux through the loop increases according to the relation $\phi_B(t) = 10t^2 + 20t$, where ϕ_B is in milliwebers and t is in seconds.

Open solution

The magnitude of current through R = 2Ω resistor at t = 5 s is _____ mA.



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Q66. Numerical

Open solution

A particle executes simple harmonic motion represented by displacement function as

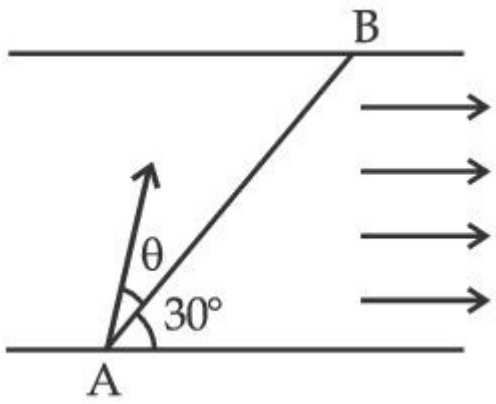
$x(t) = A \sin(\omega t + \phi)$

If the position and velocity of the particle at t = 0 s are 2 cm and 2ω cm s^{-1} respectively, then its amplitude is $x\sqrt{2}$ cm where the value of x is _____.

Q67. Numerical

Open solution

A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be _____°, so that the swimmer reaches point B.



Q68. Numerical

Open solution

For the circuit shown, the value of current at time t = 3.2 s will be _____ A.

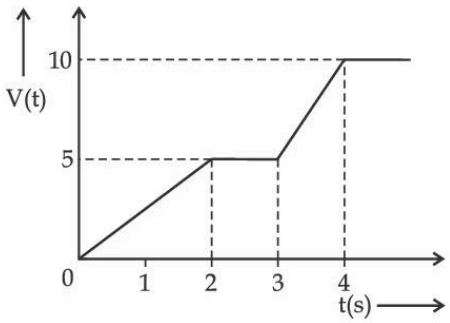


Figure 1

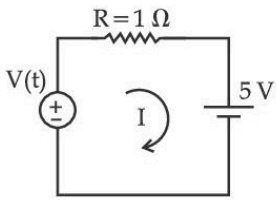


Figure 2

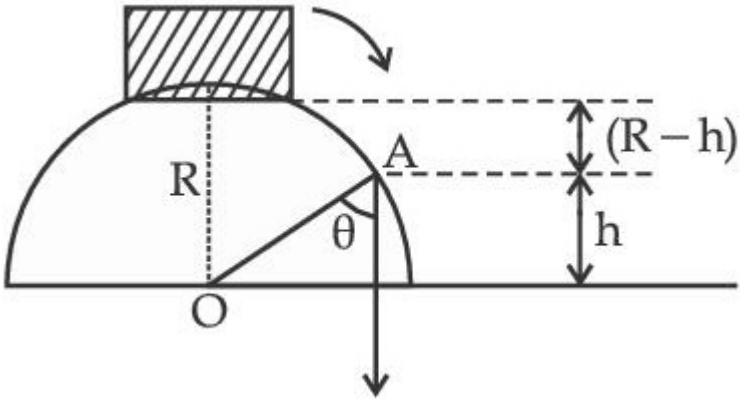
[Voltage distribution V(t) is shown by Fig. (1) and the circuit is shown in Fig. (2)]

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Q69. Numerical

A small block slides down from the top of hemisphere of radius R = 3 m as shown in the figure. The height 'h' at which the block will lose contact with the surface of the sphere is _____ m.

(Assume there is no friction between the block and the hemisphere)



Open solution

Q70. Numerical

The K_α X-ray of molybdenum has wavelength 0.071 nm. If the energy of a molybdenum atom with a K electron knocked out is 27.5 keV, the energy of this atom when an L electron is knocked out will be _____ keV. (Round off to the nearest integer)

[h = 4.14×10^{-15} eVs, c = 3×10^{8} ms^{-1}]

Open solution

Q71. Numerical

The water is filled upto height of 12 m in a tank having vertical sidewalls. A hole is made in one of the walls at a depth 'h' below the water level. The value of 'h' for which the emerging steam of water strikes the ground at the maximum range is _____ m.

Open solution

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

Which one of the following sets of elements can be detected using sodium fusion extract?

(A) Sulfur, Nitrogen, Phosphorus, Halogens

(B) Phosphorus, Oxygen, Nitrogen, Halogens

(C) Nitrogen, Phosphorus, Carbon, Sulfur

(D) Halogens, Nitrogen, Oxygen, Sulfur

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The number of neutrons and electrons, respectively, present in the radioactive isotope of hydrogen is :

(A) 1 and 1 **(B)** 3 and 1 **(C)** 2 and 1 **(D)** 2 and 2

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

	List - I		List - II
(a)	Li	(i)	photoelectric cell
(b)	Na	(ii)	absorbent of CO_2
(c)	K	(iii)	coolant in fast breeder nuclear reactor
(d)	Cs	(iv)	treatment of cancer
		(v)	bearings for motor engines

Open solution

Choose the correct answer from the options given below :

(A) (a) - (v), (b) - (i), (c) - (ii), (d) - (iv)

(B) (a) - (v), (b) - (ii), (c) - (iv), (d) - (i)

(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (v), (b) - (iii), (c) - (ii), (d) - (i)

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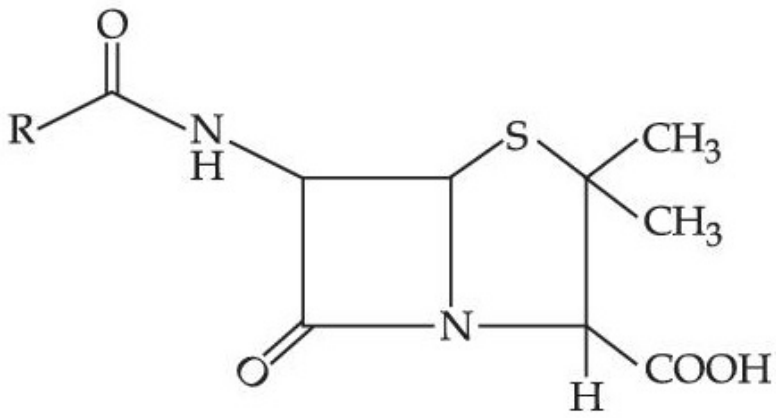
Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : Penicillin is a bacteriostatic type antibiotic.

Statement II : The general structure of Penicillin is :



Choose the correct option :

(A) Both statement I and statement II are false

(B) Statement I is incorrect but statement II is true

(C) Both statement I and statement II are true

(D) Statement I is correct but statement II is false

Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

If the Thomson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been :

(A) All of the α-particles pass through the gold foil without decrease in speed.

(B) α-particles are deflected over a wide range of angles.

(C) All α-particles get bounced back by 180°

(D) α-particles pass through the gold foil deflected by small angles and with reduced speed.

Q77. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are :

(A) 3, 1 and 1 (B) 4, 1 and 0 (C) 1, 1 and 3 (D) 1, 2 and 3

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Q78. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Select the correct statements

(A) Crystalline solids have long range order.

(B) Crystalline solids are isotropic

(C) Amorphous solid are sometimes called pseudo solids.

(D) Amorphous solids soften over a range of temperatures.

(E) Amorphous solids have a definite heat of fusion.

Choose the most appropriate answer from the options given below.

(A) (A), (B), (E) only
(B) (B), (D) only
(C) (C), (D) only
(D) (A), (C), (D) only

Q79. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The addition of silica during the extraction of copper from its sulphide ore :-

(A) converts copper sulphide into copper silicate
(B) converts iron oxide into iron silicate
(C) reduces copper sulphide into metallic copper
(D) reduces the melting point of the reaction mixture

Q80. MCQ (Single Correct Answer) · **Mathematics**

Open solution

For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines

$\frac{x-\alpha}{1} = \frac{y-1}{2} = \frac{z-1}{3}$ and $\frac{x-4}{\beta} = \frac{y-6}{3} = \frac{z-7}{3}$, lies on the plane x + 2y − z = 8, then $\alpha - \beta$ is equal to :

(A) 5 (B) 9 (C) 3 (D) 7

Q81. MCQ (Single Correct Answer) · **Mathematics**

Open solution

A student appeared in an examination consisting of 8 true-false type questions. The student guesses the answers with equal probability. the smallest value of n, so that the probability of guessing at least 'n' correct answers is less than $\frac{1}{2}$, is :

(A) 5 (B) 6 (C) 3 (D) 4

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Q82. MCQ (Single Correct Answer) · **Mathematics**

Which of the following is the negation of the statement "for all M > 0, there exists x∈S such that x $\geq$ M" ?

(A) there exists M > 0, such that x < M for all x∈S

(B) there exists M > 0, there exists x∈S such that x $\geq$ M

(C) there exists M > 0, there exists x∈S such that x < M

(D) there exists M > 0, such that x $\geq$ M for all x∈S

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $\vec{a} = \vec{b} \times (\vec{b} \times \vec{c})$. If magnitudes of the vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are $\sqrt{2}$, 1 and 2 respectively and the angle between $\vec{b}$ and $\vec{c}$ is $\theta\left(0 < \theta < \frac{\pi}{2}\right)$, then the value of 1 + tanθ is equal to :

(A) $\sqrt{3}+1$ (B) 2

(C) 1 (D) $\frac{\sqrt{3}+1}{\sqrt{3}}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let A and B be two 3 $\times$ 3 real matrices such that ($A^2 - B^2$) is invertible matrix. If $A^5 = B^5$ and $A^3B^2 = A^2B^3$, then the value of the determinant of the matrix $A^3 + B^3$ is equal to :

(A) 2 (B) 4 (C) 1 (D) 0

Open solution

Q85. Numerical · **Mathematics**

Let $\vec{a} = \hat{i} - \alpha\hat{j} + \beta\hat{k}$, $\vec{b} = 3\hat{i} + \beta\hat{j} - \alpha\hat{k}$ and $\vec{c} = -\alpha\hat{i} - 2\hat{j} + \hat{k}$, where α and β are integers. If $\vec{a}.\vec{b} = -1$ and $\vec{b}.\vec{c} = 10$, then $\left(\vec{a} \times \vec{b}\right).\vec{c}$ is equal to _____.

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Q86. Numerical · **Mathematics**

The distance of the point P(3, 4, 4) from the point of intersection of the line joining the points Q(3, -4, -5) and R(2, -3, 1) and the plane 2x + y + z = 7, is equal to _____.

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Q87. Numerical · **Mathematics**

Let E be an ellipse whose axes are parallel to the co-ordinates axes, having its center at (3, -4), one focus at (4, -4) and one vertex at (5, -4). If mx $-$ y = 4, m > 0 is a tangent to the ellipse E, then the value of $5m^2$ is equal to _____.

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Q88. MCQ (Single Correct Answer) · **Physics**

Two Carnot engines A and B operate in series such that engine A absorbs heat at T_1 and rejects heat to a sink at temperature T. Engine B absorbs half of the heat rejected by Engine A and rejects heat to the sink at T_3. When workdone in both the cases is equal, to value of T is :

Open solution

(A) $\frac{2}{3}T_1 + \frac{3}{2}T_3$

(B) $\frac{1}{3}T_1 + \frac{2}{3}T_3$

(C) $\frac{3}{2}T_1 + \frac{1}{3}T_3$

(D) $\frac{2}{3}T_1 + \frac{1}{3}T_3$

Q89. MCQ (Single Correct Answer) · **Physics**

Consider the following statements :

Open solution

A. Atoms of each element emit characteristics spectrum.

B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.

C. The density of nuclear matter depends on the size of the nucleus.

D. A free neutron is stable but a free proton decay is possible.

E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below :

(A) A, B, C, D and E (B) A, B and E only (C) B and D only (D) A, C and E only

Q90. Numerical · **Physics**

The maximum amplitude for an amplitude modulated wave is found to be 12V while the minimum amplitude is found to be 3V. The modulation index is 0.6x where x is _____.

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Answer Key

Chemistry

1. **B**	2. **B**	3. **C**	4. **C**	5. **D**
6. **A**	7. **D**	8. **C**	9. **C**	10. **A**
11. **D**	12. **C**	13. **16**	14. **82**	15. **875**
16. **1575**	17. **10**	18. **6**	19. **125**	20. **60**
21. **600 to 700**	22. **28**			

Mathematics

23. **B**	24. **D**	25. **C**	26. **D**	27. **C**
28. **C**	29. **C**	30. **B**	31. **C**	32. **B**
33. **A**	34. **B**	35. **B**	36. **B**	37. **C**
38. **1**	39. **5**	40. **2**	41. **2**	42. **924**
43. **832**	44. **2020**			

Physics

45. **B**	46. **B**	47. **C**	48. **B**	49. **A**
50. **A**	51. **B**	52. **D**	53. **C**	54. **B**
55. **C**	56. **B**	57. **B**	58. **C**	59. **D**
60. **D**	61. **C**	62. **C**	63. **9**	64. **2**
65. **60**	66. **2**	67. **30**	68. **1**	69. **2**
70. **10**	71. **6**			

Out of syllabus

72. **A**	73. **C**	74. **D**	75. **B**	76. **D**
77. **D**	78. **D**	79. **B**	80. **D**	81. **A**
82. **A**	83. **B**	84. **D**	85. **9**	86. **7**
87. **3**	88. **D**	89. **B**	90. **1**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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